Supplementary Material for A Self-Calibrating Single-Camera Pressure Controller for Heterogeneous Urban Traffic

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Abstract

This supplement holds the parameter table and the per-experiment figures and tables that the main text summarises in prose. Experiment numbers, symbols and macros are those of the main text, and every number is generated from the same raw result files by the same scripts.

I. Controller parameters

TABLE S1
CONTROLLER PARAMETERS. THE SINGLE-JUNCTION COLUMN IS USED IN EXPERIMENTS 1, 3, 6, 7, 8 AND 10, WITH $g_{\max} = 30$ S IN EXPERIMENTS 1 AND 3 AND 45 S IN THE OTHERS; THE CORRIDOR COLUMN IS THE CONFIGURATION OF THE SYSTEM'S NETWORK DEMONSTRATION, USED IN EXPERIMENTS 2 AND 9. EQUATION NUMBERS REFER TO THE MAIN TEXT.

Symbol	Meaning	Single	Corridor
$g_{\min}$	minimum green (s)	5.0	5.0
$g_{\max}$	maximum green (s)	30 / 45	34.0
$C_{\max}$	cycle clamp for the green floor (s)	120	120
τ	estimator time constant (s)	60.0	60.0
y	yellow (s)	3.0	2.0
ρ	all-red clearance (s)	1.5	2.5
γ_t	gap-out time (s)	2.5	2.5
R	vehicle maximum wait (s)	60.0	26.0
V	pedestrian maximum wait (s)	60.0	60.0
W	walk-hold allowance (s)	12.0	12.0
Θ	preemption cap (s)	60.0	25.0
θ	emergency latch (s)	2.5	2.5
h	hysteresis margin (PCU)	1.0	1.0
w_{ped}	pedestrian weight	0.3	0.3
γ	downstream coupling	n/a	0.4

II. Single junction

TABLE S2
EXPERIMENT 1. MEAN DELAY PER VEHICLE (S, STANDARD DEVIATION ACROSS 30 SEEDS IN PARENTHESES) AND THE PAIRED PER-SEED REDUCTION ACHIEVED BY RELAY AGAINST EACH FIXED-TIME BASELINE. ALL DIFFERENCES ARE SIGNIFICANT AT $p < 0.05$ BY THE WILCOXON SIGNED-RANK TEST (LARGEST $p = 0$).

	Demand	Mean delay per vehicle (s)			Reduction (%)	
Profile	(veh/h)	Equal	Webster	RELAY	vs equal	vs Webster
Asymmetric	767	11.0 (0.6)	8.4 (0.6)	5.0 (0.3)	+54.6 (3.8)	+40.1 (5.0)
	1022	11.8 (0.7)	8.9 (0.6)	5.5 (0.3)	+53.0 (4.1)	+37.7 (5.0)
	1278	13.9 (0.8)	9.6 (0.4)	6.4 (0.3)	+54.0 (4.5)	+33.3 (4.4)
	1534	19.2 (3.1)	10.7 (0.8)	7.2 (0.4)	+61.7 (6.2)	+32.5 (4.7)
	1789	45.1 (17.2)	11.8 (1.0)	8.1 (0.4)	+79.7 (6.8)	+31.3 (3.8)
Balanced	769	10.5 (0.6)	9.9 (0.4)	6.1 (0.3)	+42.0 (3.3)	+38.5 (3.0)
	1025	10.9 (0.4)	10.5 (0.4)	6.8 (0.3)	+38.0 (3.9)	+35.9 (3.4)
	1282	11.4 (0.3)	11.2 (0.4)	7.6 (0.3)	+33.4 (3.1)	+32.0 (2.9)
	1538	12.1 (0.6)	11.9 (0.6)	8.5 (0.4)	+29.7 (4.0)	+28.2 (3.7)
	1794	13.2 (0.6)	13.5 (0.6)	9.6 (0.4)	+27.4 (2.8)	+28.7 (3.1)
Single	767	14.4 (2.2)	8.8 (0.5)	5.4 (0.4)	+62.5 (4.0)	+39.3 (5.2)
	1022	38.3 (16.6)	9.8 (0.5)	6.5 (0.4)	+79.9 (8.2)	+34.0 (5.8)
	1278	151.8 (30.9)	11.8 (0.9)	7.7 (0.4)	+94.8 (1.0)	+34.7 (3.9)
	1534	239.8 (28.3)	17.1 (3.0)	9.4 (0.8)	+96.0 (0.5)	+43.8 (8.7)
	1789	307.2 (23.7)	21.3 (4.1)	12.8 (1.7)	+95.8 (0.5)	+38.7 (9.9)

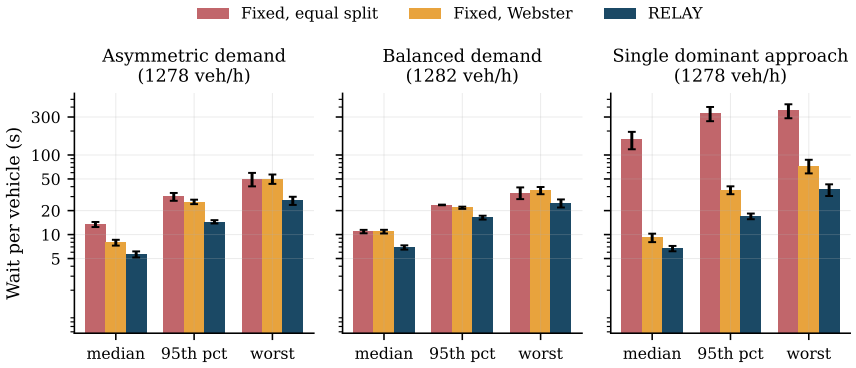


Fig. S1. Experiment 1. Median, 95th-percentile and worst observed wait per vehicle at demand multiplier 1.0, logarithmic axis. The tail is where the anti-starvation bound shows.

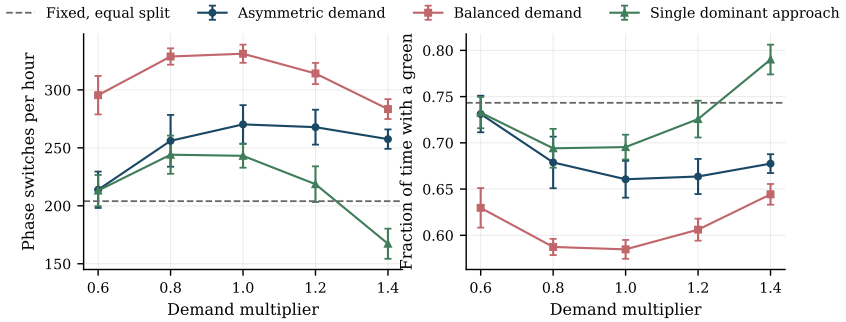


Fig. S2. Experiment 1b. Left: phase changes per hour. Right: fraction of wall time with a green displayed; the complement is clearance. Dashed: the equal-split plan, whose behaviour is independent of demand.

III. Point-queue corridor

TABLE S3
EXPERIMENT 2. THREE-JUNCTION CORRIDOR, MEAN OVER 20 SEEDS (SD IN PARENTHESES). DELAY IS TOTAL QUEUEING DELAY PER VEHICLE ACROSS THE JUNCTIONS TRAVERSED; THROUGH TRAVEL TIME IS FOR THE TWO ARTERIAL THROUGH ROUTES (FREE-FLOW 21.6 S); EMERGENCY TRAVERSAL IS THE SINGLE EMERGENCY VEHICLE PER RUN.

Demand (veh/h)	Policy	Mean delay per vehicle (s)	Through travel (s)	Stops per vehicle	Emergency traversal (s)
769	Webster, independent	26.7 (0.9)	39.7 (0.9)	1.95	36.0 (0.0)
	Webster, coordinated	25.7 (0.7)	37.0 (0.7)	1.78	24.5 (0.0)
	RELAY, no coupling	11.3 (0.5)	24.9 (0.5)	1.09	22.3 (3.3)
	RELAY, coupled	11.2 (0.5)	24.9 (0.5)	1.09	22.3 (3.3)
1282	Webster, independent	30.1 (0.8)	41.5 (0.7)	2.09	53.6 (7.5)
	Webster, coordinated	29.2 (1.1)	38.0 (0.8)	1.80	31.5 (0.0)
	RELAY, no coupling	15.2 (0.7)	28.3 (0.6)	1.53	23.7 (4.7)
	RELAY, coupled	15.2 (0.6)	28.5 (0.6)	1.53	23.5 (4.7)
1794	Webster, independent	36.7 (3.3)	43.1 (1.3)	2.10	30.1 (6.5)
	Webster, coordinated	35.3 (3.6)	39.3 (0.9)	2.03	24.0 (0.0)
	RELAY, no coupling	20.2 (0.8)	32.1 (0.7)	1.87	27.6 (8.5)
	RELAY, coupled	20.2 (0.8)	32.3 (0.7)	1.87	26.7 (8.3)

IV. Preemption latency

TABLE S4
EXPERIMENT 3. DELAY FROM FIRST EMERGENCY DETECTION TO GREEN ON THE EMERGENCY APPROACH OVER 900 TRIALS. THE EQUAL-SPLIT BASELINE HAS NO PREEMPTION PATH; ITS LATENCY IS SET BY CYCLE POSITION AND IS POOLED.

Demand	Policy	Mean	SD	Median	95th pct	Worst
Light (m=0.6)	RELAY	4.02	3.56	4.59	10.80	13.61
Medium (m=1.0)	RELAY	4.04	3.88	4.27	11.83	13.91
Heavy (m=1.4)	RELAY	4.16	3.86	4.53	11.85	13.99
All	RELAY	4.07	3.76	4.53	11.33	13.99
All	Equal split	7.00	7.04	5.02	20.59	22.29

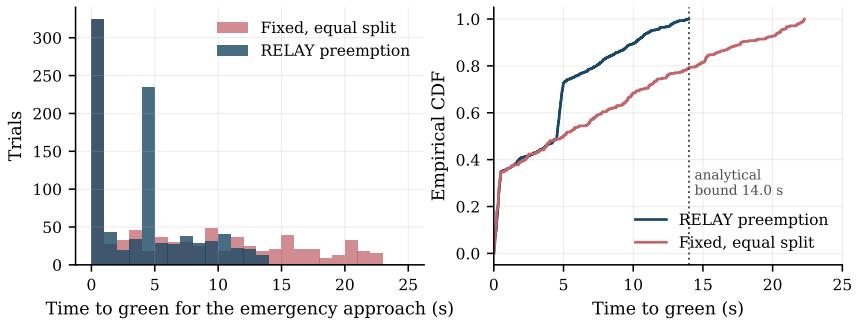


Fig. S3. Experiment 3. Left: time from first emergency detection to green, 900 trials. Right: empirical cumulative distributions against the closed-form bound of 14.0 s.

V. Perception

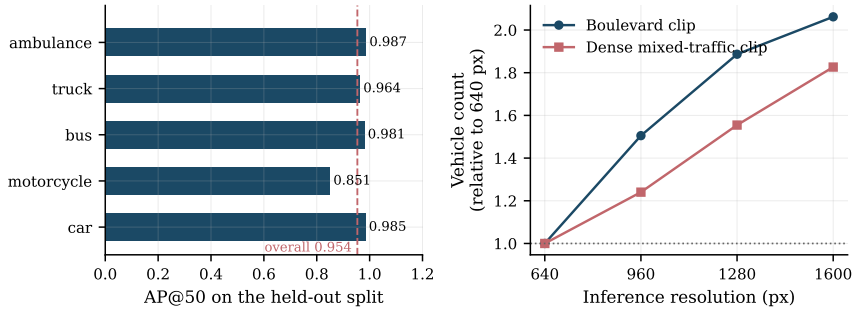


Fig. S4. Left: Experiment 4, per-class AP at IoU 0.5. Right: Experiment 5, sampled count against inference resolution, normalised to 640 px. Neither curve has flattened by 1600 px.

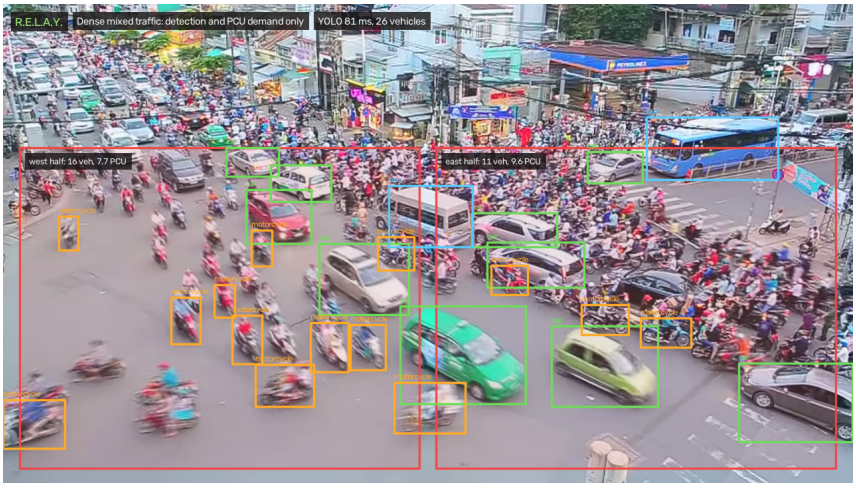


Fig. S5. Experiment 5. The stack on dense motorcycle-dominated footage. The failure mode is visible: a few dozen vehicles are detected in a scene containing several hundred, so the reported PCU demand is a severe underestimate. This is the error Experiment 6b models.

VI. BIASED-COUNT GRID

TABLE S5
EXPERIMENT 6B. THE 72-CELL GRID UNDER BIASED COUNT MODELS APPLIED TO WHAT THE CONTROLLER SEES. WON COUNTS CELLS WITH A POSITIVE REDUCTION AGAINST BOTH FIXED PLANS; WORST AND MEAN ARE THE PAIRED REDUCTION AGAINST WEBSTER OVER THE GRID.

Count model	Won	Worst (%)	Mean (%)
Saturating, $q_{1/2} = 8$	71/72	-8.9	$+36.0$
Saturating, $q_{1/2} = 4$	71/72	-22.1	$+35.3$
Scale 0.8	71/72	-0.0	$+36.6$
Scale 0.6	72/72	$+0.9$	$+36.6$
Scale 0.4 (lone vehicle rounds to 0)	51/72	-36.6	$+3.9$
Scale 0.4, presence floor 1	72/72	$+0.4$	$+36.4$
Scale 0.8, lone vehicle reads 0	50/72	-36.6	$+3.8$
Scale 0.6, lone vehicle reads 0	50/72	-36.6	$+3.9$
Scale 0.8, presence floor 1	71/72	-0.0	$+36.6$
Scale 0.6, presence floor 1	72/72	$+0.9$	$+36.6$

VII. SUMO SINGLE JUNCTION

TABLE S6
EXPERIMENT 8. SUMO, MEAN TIME LOSS PER VEHICLE (S, SD OVER 10 SEEDS) AND PAIRED REDUCTION OF RELAY AGAINST EACH BASELINE.

	Demand	Mean time loss (s)				Reduction (%)		
Profile	(veh/h)	Webster	Max-pr.	Actuated	RELAY	vs Web.	vs M-p.	vs Act.
Asymmetric	767	7.6 (0.4)	5.2 (0.4)	7.7 (0.4)	3.5 (0.2)	$+53.0$	$+31.8$	$+53.7$
	1278	8.2 (0.5)	7.0 (0.4)	8.4 (0.4)	4.8 (0.3)	$+40.9$	$+30.5$	$+42.6$
	1789	8.7 (0.4)	8.3 (0.5)	9.0 (0.6)	6.5 (0.5)	$+25.5$	$+21.8$	$+28.4$
Balanced	769	9.2 (0.3)	6.4 (0.4)	7.8 (0.3)	4.1 (0.3)	$+55.5$	$+36.1$	$+47.2$
	1282	9.7 (0.4)	8.4 (0.3)	8.7 (0.3)	5.8 (0.3)	$+40.1$	$+30.8$	$+33.0$
	1794	10.5 (0.4)	10.0 (0.9)	9.5 (0.6)	7.8 (0.4)	$+25.8$	$+21.5$	$+18.1$
Single	767	7.7 (0.4)	5.1 (0.5)	8.1 (0.3)	3.4 (0.2)	$+55.5$	$+32.4$	$+57.4$
	1278	8.4 (0.4)	7.1 (0.5)	8.8 (0.4)	5.1 (0.3)	$+39.1$	$+28.4$	$+42.6$
	1789	8.9 (0.3)	8.8 (0.5)	9.6 (0.5)	6.9 (0.5)	$+22.9$	$+22.3$	$+28.5$

VIII. SUMO CORRIDOR

TABLE S7
EXPERIMENT 9. SUMO CORRIDOR WITH 150 M LINKS, MEAN TIME LOSS PER VEHICLE (S, SD OVER 10 SEEDS) AND ARTERIAL THROUGH TRAVEL TIME (S).

Demand (veh/h)	Policy	Mean time loss (s)	Through travel (s)
769	Webster, independent	22.2 (1.0)	93.3
	Webster, coordinated	20.8 (0.7)	91.0
	RELAY, no coupling	6.9 (0.5)	78.7
	RELAY, coupled	6.7 (0.5)	78.5
1282	Webster, independent	23.3 (0.4)	94.3
	Webster, coordinated	22.6 (0.8)	92.7
	RELAY, no coupling	10.9 (0.6)	79.9
	RELAY, coupled	10.8 (0.5)	80.0
1794	Webster, independent	26.6 (1.2)	96.0
	Webster, coordinated	26.0 (2.0)	93.5
	RELAY, no coupling	17.2 (1.8)	82.3
	RELAY, coupled	17.9 (1.7)	82.4

IX. Adaptive baselines

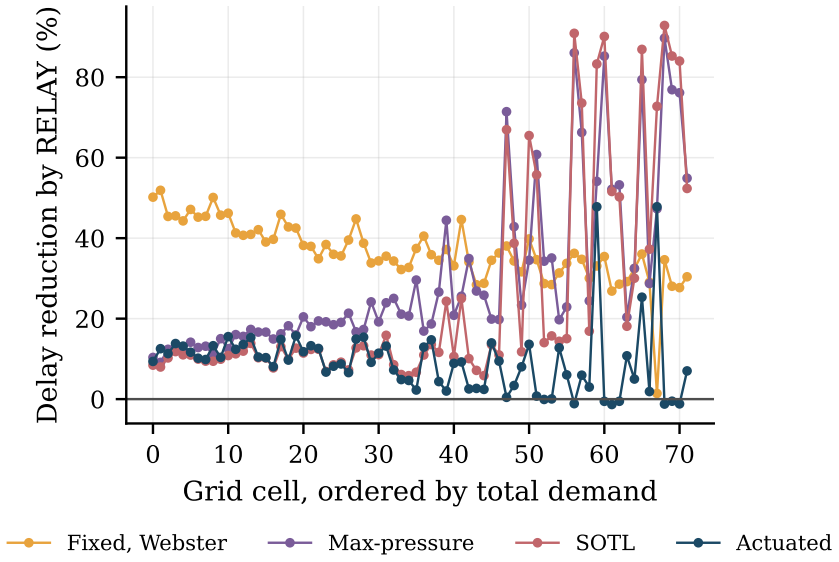


Fig. S6. Experiment 7. Paired delay reduction of RELAY against each baseline over the 72 cells, ordered by demand. Positive favours RELAY.

X. Sensitivity

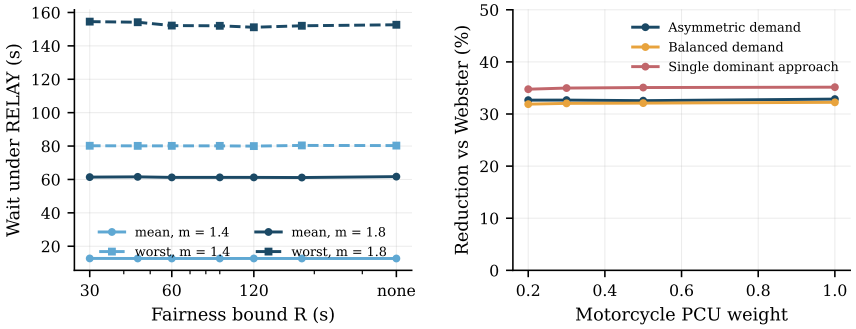


Fig. S7. Experiment 10. Left: mean and worst wait against the fairness bound R on the single-dominant shape at 1789 and 2300 veh/h. Right: reduction against Webster as a function of the motorcycle PCU weight with a two-wheeler share of 80% on one axis and 30% on the other.